

Bandwidth-Controlled Pareto Differentiation Flow: Full Derivation, Kernel Behaviour, and the Critical Bandwidth $\beta_c(\gamma)$

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Abstract

We generalize the collective Pareto differentiation flow by introducing an explicit bandwidth parameter $\beta > 0$ into the interaction kernel, $p(d) = e^{-\beta d^\gamma}$, in place of the fixed kernel e^{-d^γ} used previously. We re-derive the deterministic flow, the transverse stability tensor, and the two-target transverse eigenvalue from scratch with β kept explicit throughout. The bandwidth turns out not to be a cosmetic rescaling: it introduces a genuine second bifurcation parameter. In particular, for $\gamma > 1$ there is a critical bandwidth

$$\beta_c(\gamma) = 1 - \frac{1}{\gamma}$$

below which the terminal symmetric (Pareto) configuration is transversally *stable* rather than unstable, and a convexity argument shows this extends to a complete classification of the transverse stability profile along the entire trajectory, for every (γ, β) — summarized as an explicit four-region phase diagram in the (γ, β) plane, with each region tied to a concrete claim about *when*, in the population’s approach, splitting is first triggered. We show the terminal threshold collapses to a single combined parameter $\mu = \gamma(1 - \beta)$, universally at $\mu = 1$ — but prove this collapse is special to the terminal point and cannot be extended to the rest of the phase portrait. Adding noise to the flow, we prove a fan-width dichotomy governed by that same threshold: on the unstable side the observed spread diverges as an explicit power law in the remaining distance to the terminal point, with a source-depth- and bandwidth-dependent prefactor; on the stable side it converges to a finite stationary value that is completely *independent* of how far the population started. Both regimes are validated against direct stochastic simulation with noise injected at every integration step. We also verify that β , unlike the population size N , cannot be absorbed into a time rescaling: it moves the bifurcation set itself. Alongside the population, we give a full stability analysis of a single isolated cell ($N = 1$), including its own transverse eigenvalue, and show it obeys a clean global dichotomy independent of bandwidth: for $\gamma \geq 1$ the single-cell cost is convex, so *every* trajectory converges to the shared compromise point; for $\gamma < 1$ that point becomes a saddle and every off-axis trajectory instead reaches a single archetype exactly, in finite time.

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1 Notation

Remark 1.1 (A notational collision, resolved). The base document uses the letter b for the half-spacing between the two side targets in the three-target geometry, $\mathbf{a}_{1,2} = (\mp b, 0)$. To avoid confusion with that quantity, the new bandwidth parameter introduced here is denoted β (beta) throughout. Wherever both quantities appear together (Section 12.1), they are visually and semantically distinct: b is a *geometric* length scale of the target configuration, while β is a *kernel* parameter controlling how sharply the interaction decays with distance.

2 The bandwidth-modified kernel

2.1 Definition

We replace the interaction kernel of the base model,

$$p(d) = e^{-d^\gamma},$$

by the two-parameter family

$$p(d) = e^{-\beta d^\gamma}, \quad \beta > 0, \gamma > 0. \quad (1)$$

The exponent γ still controls the *shape* of the decay (sub-exponential for $\gamma < 1$, exponential at $\gamma = 1$, Gaussian-like at $\gamma = 2$, increasingly step-like for large γ). The new parameter β controls the *scale*: it is a bandwidth in the sense that the kernel's characteristic width shrinks as β grows. Precisely,

$$p(d) = e^{-1} \iff d = \beta^{-1/\gamma},$$

so the “e-folding radius” of the kernel is $\beta^{-1/\gamma}$: larger β means a narrower, more sharply peaked kernel; smaller β means a broader, more permissive one. The original model is recovered at $\beta = 1$.

2.2 How the kernel actually looks

Figure 1 shows $p(d) = e^{-\beta d^\gamma}$ at fixed $\gamma = 2$ for five values of β : increasing β compresses the kernel toward the origin without changing its qualitative (Gaussian) shape. Figure 2 shows the complementary picture: at fixed $\beta = 1$, changing γ changes the *shape* of the decay — from a cusped, heavy-shouldered profile at $\gamma = 0.5$, through the exponential $\gamma = 1$ and Gaussian $\gamma = 2$ cases, to an increasingly flat-then-cliff profile as γ grows.

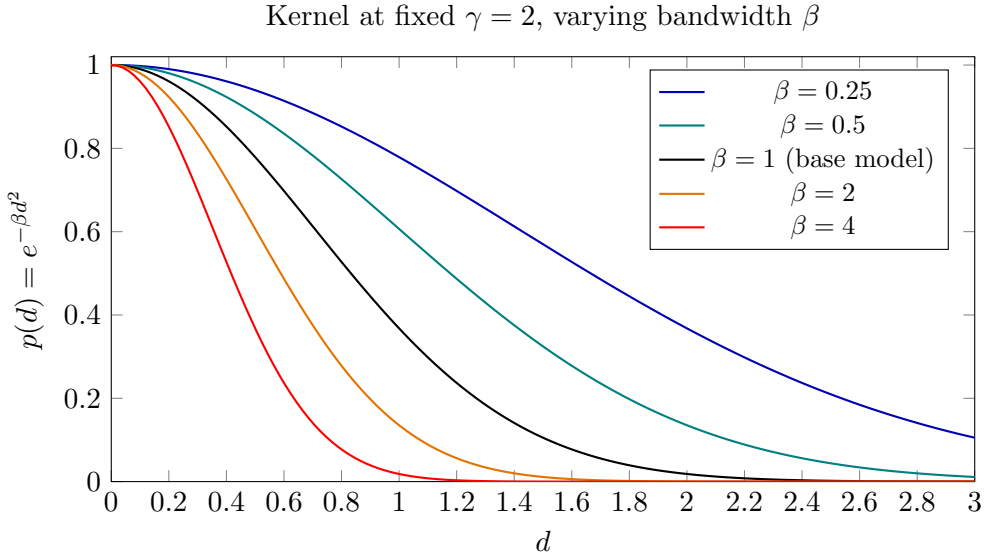


Figure 1: Increasing the bandwidth β narrows the kernel: a cell feels a target only if it is much closer, for the same exponent γ .

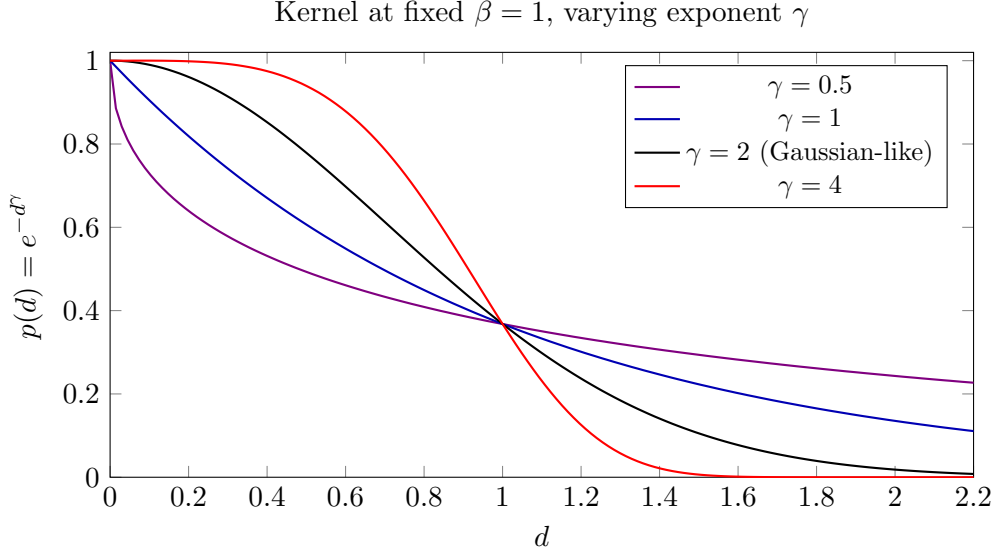


Figure 2: The exponent γ changes the *shape* of the decay, not just its scale: large γ produces an almost flat plateau followed by a sharp cliff.

Because γ and β enter the kernel jointly, the effect of increasing γ is not the same at every bandwidth. Figure 3 repeats the γ -sweep of Figure 2 at three bandwidths — broad ($\beta = 0.5$), the base model ($\beta = 1$), and narrow ($\beta = 2$) — side by side.

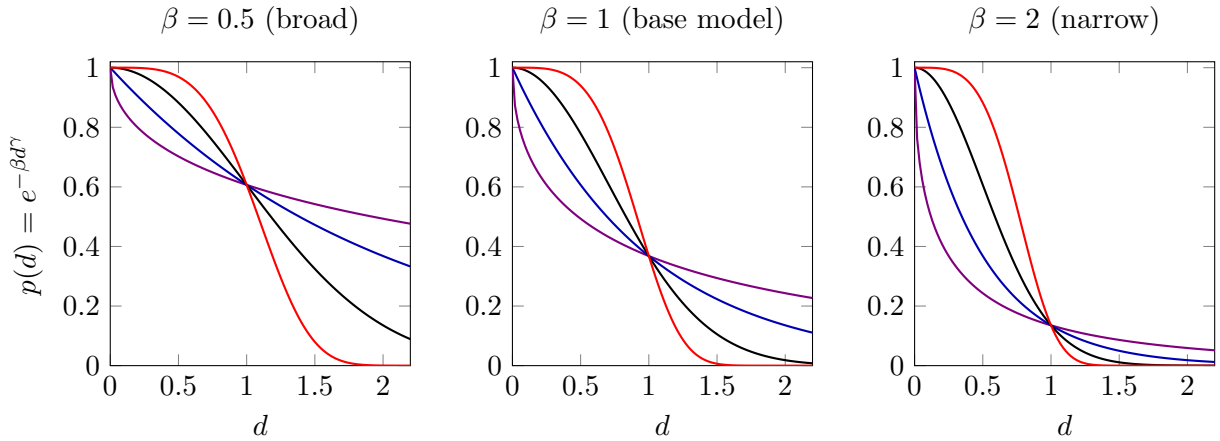


Figure 3: The effect of increasing γ under three bandwidth regimes. In every panel, larger γ still produces the same qualitative transition — from a cusped, heavy-shouldered profile toward a flat plateau with a sharp cliff — but a broad kernel ($\beta < 1$, left) keeps that cliff pushed out to larger d , while a narrow kernel ($\beta > 1$, right) pulls the entire family in toward the origin. Shape (γ) and scale (β) therefore act on the kernel independently, but their visible effect on any one curve is not: the same γ reads as a very different kernel width depending on β .

Reading the two knobs together

γ sets the *shape* of the interaction (how abruptly it turns off); β sets the *scale* at which that shape is applied. As shown below, this separation of roles is exactly what allows β to shift the location and even the character of the bifurcation without altering the underlying exponent γ .

3 The collective flow with bandwidth

The population objective is unchanged in form:

$$F(\mathbf{x}_1, \dots, \mathbf{x}_N) = \prod_{i=1}^k S_i, \quad S_i = \sum_{j=1}^N p_{ij}, \quad p_{ij} = e^{-\beta d_{ij}^\gamma}, \quad d_{ij} = \|\mathbf{x}_j - \mathbf{a}_i\|.$$

Hence

$$\log F = \sum_{i=1}^k \log S_i, \quad \dot{\mathbf{x}}_j = \nabla_{\mathbf{x}_j} \log F = \sum_{i=1}^k \frac{1}{S_i} \nabla_{\mathbf{x}_j} p_{ij},$$

where the second equality uses $\partial S_i / \partial \mathbf{x}_j = \partial p_{ij} / \partial \mathbf{x}_j$ (only the j -th summand of S_i depends on $\mathbf{x}_j$).

3.1 Gradient of the kernel

Write $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{a}_i$. Since $\nabla d = \mathbf{r}/d$,

$$\nabla d^\gamma = \gamma d^{\gamma-1} \nabla d = \gamma d^{\gamma-2} \mathbf{r},$$

which is unchanged by the bandwidth (it is purely a property of the distance function). Therefore

$$\begin{aligned} \nabla_{\mathbf{x}_j} p_{ij} &= \nabla_{\mathbf{x}_j} \exp(-\beta d_{ij}^\gamma) \\ &= -\beta (\nabla d_{ij}^\gamma) e^{-\beta d_{ij}^\gamma} \\ &= -\beta \gamma d_{ij}^{\gamma-2} p_{ij} \mathbf{r}_{ij}. \end{aligned} \tag{2}$$

3.2 The exact flow

Substituting (2) into $\dot{\mathbf{x}}_j$:

$$\dot{\mathbf{x}}_j = -\beta \gamma \sum_{i=1}^k \frac{p_{ij}}{S_i} d_{ij}^{\gamma-2} (\mathbf{x}_j - \mathbf{a}_i). \tag{3}$$

This is identical in form to the base-model flow, with the single replacement $\gamma \mapsto \beta \gamma$ in the *overall prefactor* — but, as the next sections show, β also enters *inside* every stability eigenvalue, not merely as an overall rate.

3.3 The single-cell limit: performance as a product across tasks

It is worth pausing on the special case $N = 1$ before returning to the population, since it is what actually justifies the multiplicative form of F in the first place.

Definition 3.1 (Performance on a task). The performance of a cell at position $\mathbf{x}$ on task (target) i is the kernel value itself,

$$\pi_i(\mathbf{x}) := e^{-\beta d_i^\gamma}, \quad d_i = \|\mathbf{x} - \mathbf{a}_i\|. \tag{4}$$

It equals 1 exactly at the archetype $\mathbf{a}_i$ and decays toward 0 as the cell moves away, at the rate set by (β, γ) studied in Section 2.

Proposition 3.2 (Single-cell objective is a product of task performances). *Let m denote the number of tasks (this is the quantity called k elsewhere in this document). For $N = 1$,*

$$F(\mathbf{x}) \Big|_{N=1} = \prod_{i=1}^m \pi_i(\mathbf{x}). \tag{5}$$

Proof. With a single cell, every per-task sum has exactly one summand:

$$S_i = \sum_{j=1}^N p_{ij} \Big|_{N=1} = p_{i1} = e^{-\beta d_{i1}^\gamma} = \pi_i(\mathbf{x}).$$

Since $F = \prod_{i=1}^m S_i$ by definition (Section 3), substituting gives $F|_{N=1} = \prod_{i=1}^m \pi_i(\mathbf{x})$ directly. $\square$

Remark 3.3. Proposition 3.2 shows that the population-level rule — multiply the per-task totals S_i across tasks — is exactly the N -cell generalization of the more primitive, single-organism statement that overall fitness is the *product*, not the sum, of task performances. S_i plays, for the population, precisely the role that π_i plays for the individual cell: it is the total performance delivered on task i , still combined multiplicatively across tasks.

Corollary 3.4 (Multiplicative performance is additive cost). *Taking logarithms of (5),*

$$\log F(\mathbf{x}) \Big|_{N=1} = \sum_{i=1}^m \log \pi_i(\mathbf{x}) = -\beta \sum_{i=1}^m d_i^\gamma. \quad (6)$$

Hence maximizing the multiplicative performance $\prod_i \pi_i(\mathbf{x})$ is exactly equivalent to minimizing the additive, γ -weighted total cost $C(\mathbf{x}) = \beta \sum_{i=1}^m d_i^\gamma$: multiplicative performance and additive cost are the same object viewed on two different scales.

Why a product, not a sum, of performances

The product structure encodes an *AND* across tasks rather than an *OR*. If a cell fails one task entirely, $\pi_i \rightarrow 0$, then $F = \prod_i \pi_i \rightarrow 0$ regardless of how well it performs on every other task — one weak link collapses total fitness. An additive combination $\sum_i \pi_i$ would instead let excellence on one task compensate for failure on another. It is this all-or-nothing requirement — good at *every* task simultaneously, not merely on average — that gives the model its Pareto character, and it is ultimately what forces a *population* to specialize (Sections 5–8) rather than settle everyone at a single shared compromise.

Proposition 3.5 (Single-cell dynamics). *The gradient flow of an isolated cell is*

$$\dot{\mathbf{x}} = \nabla \log F(\mathbf{x}) \Big|_{N=1} = -\beta \gamma \sum_{i=1}^m d_i^{\gamma-2} (\mathbf{x} - \mathbf{a}_i). \quad (7)$$

Proof. Set $N = 1$ in (3). With a single cell, $S_i = p_{i1}$ exactly, so the ratio appearing there is identically

$$\frac{p_{i1}}{S_i} = \frac{p_{i1}}{p_{i1}} = 1$$

for the (only) cell $j = 1$. Substituting into (3) gives (7) directly. Equivalently, this is just $\dot{\mathbf{x}} = -\nabla C(\mathbf{x})$ for the cost of Corollary 3.4. $\square$

Proposition 3.6 (Single-cell stability tensor). *Let $J^{(1)}(\mathbf{x}) = D_{\mathbf{x}} \dot{\mathbf{x}} = \nabla^2 \log F(\mathbf{x})|_{N=1}$ be the Jacobian of the single-cell flow (7) at any point $\mathbf{x}$ (not necessarily an equilibrium) — i.e. the ordinary $D \times D$ matrix of partial derivatives of the vector field $\dot{\mathbf{x}} = f(\mathbf{x})$ on $\mathbb{R}^D$ ($D = 2$ in the planar examples below). This is the same word used for the collective Jacobian J_{jk} of Section 5, but a much simpler object: there is only one cell, so there are no cell indices j, k to couple, and $J^{(1)}$ is literally the Hessian of $\log F$ rather than a block matrix built from it. Then*

$$J^{(1)}(\mathbf{x}) = -\beta \gamma \sum_{i=1}^m d_i^{\gamma-2} \left[I + (\gamma - 2) \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^\top \right], \quad (8)$$

and for a unit direction $\mathbf{e}$, the directional eigenvalue is

$$\lambda^{(1)}(\mathbf{e}; \mathbf{x}) = \mathbf{e}^\top J^{(1)}(\mathbf{x}) \mathbf{e} = -\beta\gamma \sum_{i=1}^m d_i^{\gamma-2} \left[1 + (\gamma-2)(\mathbf{e} \cdot \hat{\mathbf{r}}_i)^2 \right]. \quad (9)$$

At any equilibrium $\mathbf{x}^*$ of (7), $\mathbf{x}^*$ is locally stable in direction $\mathbf{e}$ iff $\lambda^{(1)}(\mathbf{e}; \mathbf{x}^*) < 0$, and unstable iff $\lambda^{(1)}(\mathbf{e}; \mathbf{x}^*) > 0$, by the usual linearization $\delta \dot{\mathbf{x}} = J^{(1)}(\mathbf{x}^*) \delta \mathbf{x}$.

Proof. Differentiate (7) once more. Writing $g_i(\mathbf{x}) = d_i^{\gamma-2} \mathbf{r}_i$ (so that $\dot{\mathbf{x}} = -\beta\gamma \sum_i g_i$), the product rule gives

$$\nabla g_i = (\nabla d_i^{\gamma-2}) \mathbf{r}_i^\top + d_i^{\gamma-2} \nabla \mathbf{r}_i = (\gamma-2) d_i^{\gamma-4} \mathbf{r}_i \mathbf{r}_i^\top + d_i^{\gamma-2} I,$$

using $\nabla d_i^{\gamma-2} = (\gamma-2) d_i^{\gamma-4} \mathbf{r}_i$ (the same identity as $\nabla d^\gamma = \gamma d^{\gamma-2} \mathbf{r}$, with exponent $\gamma-2$ in place of γ) and $\nabla \mathbf{r}_i = I$. Since $\mathbf{r}_i \mathbf{r}_i^\top = d_i^2 \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^\top$,

$$\nabla g_i = d_i^{\gamma-2} \left[I + (\gamma-2) \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^\top \right],$$

and summing with the prefactor $-\beta\gamma$ gives (8). Equation (9) follows by contracting with $\mathbf{e}$ on both sides. $\square$

Remark 3.7 (A structurally simpler eigenvalue than the collective case). Compare (9) with the collective directional eigenvalue (14). The collective formula contains the term $\beta\gamma d_i^\gamma$ inside the bracket — this is exactly what let β shift the location of the bifurcation in Sections 7–8. The single-cell formula (9) has no such term: the bracket $[1 + (\gamma-2)(\mathbf{e} \cdot \hat{\mathbf{r}}_i)^2]$ depends only on γ and geometry, never on β . This is a direct consequence of log undoing the kernel’s exponential for a single cell (Corollary 3.4): the population’s normalizing sum S_i is what reintroduces the nonlinear $\beta\gamma d_i^\gamma$ dependence into the collective eigenvalue, and a lone cell has no such sum. Concretely, this means bandwidth can rescale the *rate* of a single cell’s stability or instability, but — unlike in the population — it can never flip its *sign*. Section 6 makes this precise on the two-target axis.

Remark 3.8 (Specialization is a collective phenomenon). Unlike the N -cell flow (3), the single-cell flow (7) has no normalizing denominator at all: there is no other cell to share task coverage with, so the entire competitive mechanism that later produces *population* symmetry breaking (Sections 5–8) is simply absent. This does not mean a single cell’s own trajectory is always stable, however — Proposition 3.6 shows it has genuine transverse eigenvalues of its own, and Section 6 finds a regime where they are positive. What is absent for $N = 1$ is specifically the mechanism by which a *population* splits into distinct sub-groups; an unstable single cell simply drifts off its symmetric path toward one target, which is not yet “specialization” in the collective sense, only the loss of a symmetric trajectory. For $m \geq 2$ distinct targets, an isolated cell instead settles — when it settles — at a single compromise point minimizing $C(\mathbf{x}) = \beta \sum_i d_i^\gamma$ across all targets simultaneously.

4 Two symmetric targets: axial motion

Take $\mathbf{a}_1 = (-1, 0)$, $\mathbf{a}_2 = (1, 0)$ and a source on the symmetry axis, $\mathbf{x}_j(t) = (0, y(t))$ for every j . Both targets are equidistant:

$$d(y) = \sqrt{1 + y^2}.$$

On the axis, $S_1 = S_2 = N e^{-\beta d^\gamma}$, so

$$\log F = 2 \log N - 2\beta d^\gamma.$$

Differentiating with respect to y and dividing by N (each of the N cells contributes the same axial pull, and S_i already carries the factor of N):

$$\boxed{\dot{y} = -\kappa(y) y, \quad \kappa(y) = \frac{2\beta\gamma}{N}(1+y^2)^{(\gamma-2)/2} > 0.} \quad (10)$$

Near $y = 0$, $\dot{y} \approx -\frac{2\beta\gamma}{N}y$, so

$$y(t) \sim y_0 e^{-2\beta\gamma t/N}.$$

The axial approach to the symmetric terminal point is therefore still monotone and asymptotic for any $\beta > 0$; increasing β (or γ) simply speeds up this approach along the axis. This part of the picture is qualitatively unchanged from the base model — the interesting effect of β appears entirely in the *transverse* direction.

5 Transverse stability: full re-derivation

5.1 Hessian of the bandwidth-modified kernel

Let $\phi = \beta d^\gamma$, so $p = e^{-\phi}$. Using the standard identity $\nabla^2 e^{-\phi} = e^{-\phi}[(\nabla\phi)(\nabla\phi)^\top - \nabla^2\phi]$, we need $\nabla\phi$ and $\nabla^2\phi$.

Since $\nabla d^\gamma = \gamma d^{\gamma-2}\mathbf{r}$,

$$\nabla\phi = \beta\gamma d^{\gamma-2}\mathbf{r}.$$

For the Hessian of d^γ itself (unchanged from the base model's Appendix A, since it does not involve β):

$$\nabla^2 d^\gamma = \gamma \left[d^{\gamma-2}I + (\gamma-2)d^{\gamma-4}\mathbf{r}\mathbf{r}^\top \right],$$

so

$$\nabla^2\phi = \beta\gamma \left[d^{\gamma-2}I + (\gamma-2)d^{\gamma-4}\mathbf{r}\mathbf{r}^\top \right].$$

Also

$$(\nabla\phi)(\nabla\phi)^\top = \beta^2\gamma^2 d^{2\gamma-4}\mathbf{r}\mathbf{r}^\top.$$

Combining,

$$\begin{aligned} \nabla^2 p &= e^{-\phi} \left[\beta^2\gamma^2 d^{2\gamma-4}\mathbf{r}\mathbf{r}^\top - \beta\gamma d^{\gamma-2}I - \beta\gamma(\gamma-2)d^{\gamma-4}\mathbf{r}\mathbf{r}^\top \right] \\ &= \boxed{\beta\gamma e^{-\beta d^\gamma} d^{\gamma-4} \left[(\beta\gamma d^\gamma - (\gamma-2)) \mathbf{r}\mathbf{r}^\top - d^2 I \right]}. \end{aligned} \quad (11)$$

At $\beta = 1$ this reduces exactly to the base model's kernel Hessian.

5.2 The collective transverse operator

Consider N co-located cells at a common point $\mathbf{x}$, with k targets $\mathbf{a}_i$, $d_i = \|\mathbf{x} - \mathbf{a}_i\|$, $\hat{\mathbf{r}}_i = (\mathbf{x} - \mathbf{a}_i)/d_i$. At this configuration $S_i = Np_i(\mathbf{x})$.

Writing the flow as $\dot{\mathbf{x}}_j = \sum_i g_i(\mathbf{x}_j)/S_i$ with $g_i = \nabla p_i$, and linearising about the co-located point, the Jacobian block coupling cells j and k is

$$J_{jk} = H \delta_{jk} - Q, \quad H = \sum_i \frac{\nabla^2 p_i}{S_i} = \frac{1}{N} \sum_i \frac{\nabla^2 p_i}{p_i}, \quad Q = \sum_i \frac{(\nabla p_i)(\nabla p_i)^\top}{S_i^2}.$$

For a *zero-sum* perturbation $\sum_j \delta \mathbf{x}_j = 0$ (the population's internal, symmetry-breaking modes), the Q -term cancels identically:

$$(J \delta \mathbf{x})_j = H \delta \mathbf{x}_j - Q \sum_k \delta \mathbf{x}_k = H \delta \mathbf{x}_j.$$

So the transverse stability operator is simply H . Using (11) with $\mathbf{r}_i \mathbf{r}_i^\top = d_i^2 \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^\top$,

$$H = \frac{\beta\gamma}{N} \sum_i d_i^{\gamma-2} \left[(\beta\gamma d_i^\gamma - \gamma + 2) \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^\top - I \right] = M(\mathbf{x}) - \tau(\mathbf{x})I \equiv \Lambda_{\text{split}}(\mathbf{x}), \quad (12)$$

with

$$M(\mathbf{x}) = \frac{\beta\gamma}{N} \sum_i d_i^{\gamma-2} (\beta\gamma d_i^\gamma - \gamma + 2) \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^\top, \quad \tau(\mathbf{x}) = \frac{\beta\gamma}{N} \sum_i d_i^{\gamma-2}. \quad (13)$$

For a unit direction $\mathbf{e}$, the directional eigenvalue is

$$\lambda(\mathbf{e}) = \mathbf{e}^\top \Lambda_{\text{split}} \mathbf{e} = \frac{\beta\gamma}{N} \sum_i d_i^{\gamma-2} \left[(\beta\gamma d_i^\gamma - \gamma + 2) (\mathbf{e} \cdot \hat{\mathbf{r}}_i)^2 - 1 \right]. \quad (14)$$

Equations (13)–(14) generalize the base model's Appendix B and C by the single, uniform replacement $\gamma d_i^\gamma \mapsto \beta\gamma d_i^\gamma$ inside every bracket, with an overall prefactor $\gamma \mapsto \beta\gamma$. Every other factor ($d_i^{\gamma-2}$, the $-\gamma + 2$ term, the -1) is untouched, because those come from the geometric Hessian of d^γ , not from the kernel's overall scale.

6 The two-target transverse eigenvalue

For $\mathbf{a}_{1,2} = (\mp 1, 0)$ and $\mathbf{x} = (0, y)$, both targets share $d = \sqrt{1 + y^2}$ and $(\mathbf{e}_x \cdot \hat{\mathbf{r}}_{1,2})^2 = 1/d^2$. Equation (14) with $\mathbf{e} = \mathbf{e}_x$ gives, after doubling for the two identical targets,

$$\lambda_{\text{split}} = \frac{2\beta\gamma}{N} d^{\gamma-2} \left[\frac{\beta\gamma d^\gamma - \gamma + 2}{d^2} - 1 \right] = \frac{2\beta\gamma}{N} d^{\gamma-4} [\beta\gamma d^\gamma - \gamma + 2 - d^2].$$

Introduce $u = d^2 = 1 + y^2$ and define

$$\varphi_\beta(u) = \beta\gamma u^{\gamma/2} - u - (\gamma - 2). \quad (15)$$

Then

$$\lambda_{\text{split}}(y) = \frac{2\beta\gamma}{N} (1 + y^2)^{(\gamma-4)/2} \varphi_\beta(1 + y^2). \quad (16)$$

Since the prefactor $\frac{2\beta\gamma}{N} (1 + y^2)^{(\gamma-4)/2}$ is strictly positive for all $\beta, \gamma > 0$ and all y ,

$$\text{sign}(\lambda_{\text{split}}(y)) = \text{sign}(\varphi_\beta(1 + y^2)). \quad (17)$$

This single fact is the key to everything that follows: the entire transverse-stability question reduces to the sign of the scalar function $\varphi_\beta(u)$ on $u \geq 1$.

6.1 The single-cell version, and why it is different

It is instructive to run the identical computation for a single cell ($N = 1$) on the same axis, using Proposition 3.6 in place of the collective tensor. With $\mathbf{e} = \mathbf{e}_x$ and $(\mathbf{e}_x \cdot \hat{\mathbf{r}}_{1,2})^2 = 1/d^2$, (9) gives, after doubling for the two identical targets,

$$\lambda_x^{(1)}(y) = -2\beta\gamma d^{\gamma-2} \left[1 + \frac{\gamma - 2}{d^2} \right] = -2\beta\gamma d^{\gamma-4} [d^2 + \gamma - 2]. \quad (18)$$

With $u = d^2 = 1 + y^2$ as before, define

$$\psi(u) = u + (\gamma - 2), \quad (19)$$

so that

$$\lambda_x^{(1)}(y) = -2\beta\gamma(1+y^2)^{(\gamma-4)/2}\psi(1+y^2). \quad (20)$$

The prefactor $-2\beta\gamma(1+y^2)^{(\gamma-4)/2}$ is strictly *negative* for all $\beta, \gamma > 0$, so

$$\text{sign}(\lambda_x^{(1)}(y)) = -\text{sign}(\psi(1+y^2)). \quad (21)$$

Proposition 6.1 (Single-cell terminal stability and critical height). *On the two-target axis, the single-cell terminal point ($y = 0$) satisfies*

$$\lambda_x^{(1)}(0) = -2\beta\gamma(\gamma - 1). \quad (22)$$

Consequently:

- if $\gamma > 1$: $\lambda_x^{(1)}(y) < 0$ for every y — the single cell's symmetric trajectory is transversely stable everywhere;
- if $\gamma = 1$: $\lambda_x^{(1)}(y) = 0$ at $y = 0$ and $\lambda_x^{(1)}(y) < 0$ for $y \neq 0$ — exactly marginal at the terminal point;
- if $\gamma < 1$: $\lambda_x^{(1)}(y) > 0$ for $|y| < y_c^{(1)}$ and $\lambda_x^{(1)}(y) < 0$ for $|y| > y_c^{(1)}$, with critical height

$$y_c^{(1)}(\gamma) = \sqrt{1 - \gamma}. \quad (23)$$

Proof. Since ψ is affine in u (unlike φ_β , it is neither concave nor convex — it is exactly linear), its sign on $u \geq 1$ is settled directly: $\psi(u) \geq 0 \iff u \geq 2 - \gamma$. If $\gamma \geq 1$ then $2 - \gamma \leq 1 \leq u$ always, so $\psi(u) \geq 0$ throughout, with equality only possible at $u = 1, \gamma = 1$; by (21) this gives $\lambda_x^{(1)} \leq 0$ throughout, i.e. the first two bullets. If $\gamma < 1$ then $2 - \gamma > 1$, so $\psi(u) < 0$ exactly on $1 \leq u < 2 - \gamma$, i.e. $y^2 < 1 - \gamma$, giving $\lambda_x^{(1)} > 0$ there by (21); solving $u = 2 - \gamma$ for y gives $y_c^{(1)} = \sqrt{1 - \gamma}$ exactly. $\square$

Single-cell instability is real, but structurally different from the collective one

For $\gamma < 1$, a lone cell placed exactly on the axis within $|y| < \sqrt{1 - \gamma}$ of the terminal point will, under any lateral perturbation, drift away from the axis toward one target rather than converge to the shared point — its own trajectory is transversely unstable, with no population involved at all. But compare (19) with φ_β in (15): the single-cell function $\psi(u) = u + (\gamma - 2)$ contains no β whatsoever, while $\varphi_\beta(u) = \beta\gamma u^{\gamma/2} - u - (\gamma - 2)$ does. Bandwidth therefore changes *how fast* the single cell's instability grows (through the overall prefactor $\beta\gamma$) but can never change *whether* it occurs, nor where $y_c^{(1)}$ sits — that depends on γ alone. The critical bandwidth $\beta_c(\gamma)$ of Section 7 is a genuinely collective effect, made possible only because the population's normalizing sum S_i reintroduces $\beta\gamma d_i^\gamma$ into the eigenvalue — exactly the term a single cell's log cancels away (Corollary 3.4).

	Single cell ($N = 1$)	Population (N -cell)
Stability function	$\psi(u) = u + (\gamma - 2)$, linear	$\varphi_\beta(u) = \beta\gamma u^{\gamma/2} - u - (\gamma - 2)$
Depends on β ?	no (prefactor only)	yes, inside the bracket
Terminal threshold	$\gamma = 1$	$\beta = \beta_c(\gamma) = 1 - 1/\gamma$ (for $\gamma > 1$)
Unstable region	$\gamma < 1, y < \sqrt{1 - \gamma}$	$\beta > \beta_c(\gamma)$ (or the inverted case, Sec. 8)

6.2 The concrete global picture: what actually happens for $\gamma < 1$ versus $\gamma > 1$

Proposition 6.1 is a *local* statement: it only says whether the symmetric point is stable to infinitesimal perturbations. It does not say whether that local behavior is the whole story globally, nor where an unstable trajectory actually ends up. Both questions have complete, concrete answers.

Lemma 6.2 (Convexity of the single-cell cost for $\gamma \geq 1$). *For $\gamma \geq 1$ and any targets $\mathbf{a}_1, \dots, \mathbf{a}_m \in \mathbb{R}^D$, the cost*

$$C(\mathbf{x}) = \beta \sum_{i=1}^m \|\mathbf{x} - \mathbf{a}_i\|^\gamma \quad (24)$$

is convex on $\mathbb{R}^D$, and coercive: $C(\mathbf{x}) \rightarrow \infty$ as $\|\mathbf{x}\| \rightarrow \infty$.

Proof. The scalar map $t \mapsto t^\gamma$ is convex and nondecreasing on $[0, \infty)$ for $\gamma \geq 1$, since $\frac{d}{dt}t^\gamma = \gamma t^{\gamma-1} \geq 0$ and $\frac{d^2}{dt^2}t^\gamma = \gamma(\gamma-1)t^{\gamma-2} \geq 0$. The map $\mathbf{x} \mapsto \|\mathbf{x} - \mathbf{a}_i\|$ is convex (a norm composed with a translation). The composition of a convex, nondecreasing scalar function with a convex function is convex, so each $\|\mathbf{x} - \mathbf{a}_i\|^\gamma$ is convex; C , a nonnegative sum of such terms, is therefore convex. Coercivity is immediate, since each individual summand already tends to ∞ as $\|\mathbf{x}\| \rightarrow \infty$. $\square$

Theorem 6.3 (Global convergence for $\gamma \geq 1$). *For $\gamma \geq 1$, the single-cell flow (7) (which is exactly $\dot{\mathbf{x}} = -\nabla C(\mathbf{x})$ for the cost (24), by Corollary 3.4) converges, from every initial condition, to the set of global minimizers of C . On the two-target axis this set is*

$$\arg \min C = \begin{cases} \{(0, 0)\} & \gamma > 1, \\ \{(x, 0) : -1 \leq x \leq 1\} & \gamma = 1. \end{cases} \quad (25)$$

Proof. Along any trajectory of $\dot{\mathbf{x}} = -\nabla C(\mathbf{x})$,

$$\frac{d}{dt}C(\mathbf{x}(t)) = \nabla C(\mathbf{x}(t)) \cdot \dot{\mathbf{x}}(t) = -\|\nabla C(\mathbf{x}(t))\|^2 \leq 0,$$

with equality iff $\mathbf{x}(t)$ is a critical point of C . So C decreases strictly along the flow except at critical points. By Lemma 6.2, C is coercive, so every sublevel set $\{C \leq c\}$ is closed and bounded, hence compact, and the trajectory remains in one such set for all $t \geq 0$. LaSalle's invariance principle then gives convergence to the largest invariant subset of $\{\nabla C = 0\}$, i.e. to the critical-point set of C . Since C is convex, every critical point is a global minimizer, proving the first claim. On the connecting segment $\Sigma = \{(x, 0) : -1 < x < 1\}$, where $C(x, 0) = \beta[(x+1)^\gamma + (1-x)^\gamma]$,

$$C'(x) = \beta\gamma[(x+1)^{\gamma-1} - (1-x)^{\gamma-1}],$$

which vanishes iff $(x+1)^{\gamma-1} = (1-x)^{\gamma-1}$. For $\gamma > 1$ the map $t \mapsto t^{\gamma-1}$ is injective on $(0, \infty)$, forcing $x+1 = 1-x$, i.e. $x = 0$ uniquely; for $\gamma = 1$ both sides equal 1 identically, so every $x \in (-1, 1)$ is a critical point. Off Σ , the two-target reflection symmetry leaves $(0, 0)$ as the only point where $\nabla C = 0$ can hold in the transverse direction as well (Proposition 6.1 already verifies this is a critical point for every γ), giving (25). $\square$

This upgrades a local result to a global one

For $\gamma > 1$, Proposition 6.1 only established that $(0, 0)$ is *locally* stable. Theorem 6.3 is much stronger: *every* initial condition, however far off-axis, converges to $(0, 0)$. For $\gamma > 1$ a single cell cannot specialize even in principle — not merely because nearby perturbations decay, but because the entire cost landscape has nowhere else for it to go.

For $\gamma < 1$, $t \mapsto t^\gamma$ is concave rather than convex, so Lemma 6.2 fails, and the global picture is qualitatively different: the cell does not merely fail to return to $(0, 0)$, it reaches one of the archetypes exactly, in finite time.

Theorem 6.4 (Finite-time specialization for $\gamma < 1$). *For $\gamma < 1$, on the connecting segment $\Sigma = \{(x, 0) : -1 < x < 1\}$: $x = 0$ is the unique critical point of $C|_\Sigma$, and C is strictly decreasing on $(0, 1)$ and strictly increasing on $(-1, 0)$. Consequently any cell with $y(0) = 0$, $x(0) \neq 0$ moves monotonically toward the nearer target, and reaches it in finite time: there exists $t^* < \infty$ such that*

$$\boxed{x(t) \longrightarrow \mathbf{a}_2 \text{ as } t \rightarrow t^{*-} \quad (\text{resp. } \mathbf{a}_1 \text{ if } x(0) < 0).} \quad (26)$$

Proof. Uniqueness of the critical point $x = 0$ on Σ follows from the same equation $(x+1)^{\gamma-1} = (1-x)^{\gamma-1}$ used in Theorem 6.3, whose only solution for $\gamma \neq 1$ is $x = 0$. Comparing values, $C(0, 0) = 2\beta$, while $\lim_{x \rightarrow 1^-} C(x, 0) = \beta \cdot 2^\gamma < 2\beta$ since $\gamma < 1$. As C has no other critical point on $(0, 1)$, continuity forces C to be strictly decreasing on the entire interval, so $\dot{x} = -C'(x) > 0$ throughout: the cell moves monotonically toward $x = 1$. For the finite-time claim, set $\eta = 1 - x \rightarrow 0^+$ and isolate the diverging term as $x \rightarrow 1$: the target-2 contribution to $\dot{x}$ is $-\beta\gamma\eta^{\gamma-2} \cdot \eta = -\beta\gamma\eta^{\gamma-1}$, while the target-1 contribution stays bounded (its distance tends to the fixed value 2), so $\dot{\eta} = -\dot{x} \leq -\frac{1}{2}\beta\gamma\eta^{\gamma-1}$ once η is small enough. Separating variables,

$$t^* - t_0 \leq \frac{2}{\beta\gamma} \int_0^{\eta(t_0)} \eta^{1-\gamma} d\eta = \frac{2\eta(t_0)^{2-\gamma}}{\beta\gamma(2-\gamma)} < \infty,$$

since $2 - \gamma > 0$ for $\gamma < 1$. Hence $\eta \rightarrow 0$, i.e. $\mathbf{x} \rightarrow \mathbf{a}_2$, in finite time. The case $x(0) < 0$ is identical with $\mathbf{a}_1$ in place of $\mathbf{a}_2$. $\square$

The concrete dichotomy

$\gamma \geq 1$: every trajectory converges (asymptotically) to the single shared compromise point.

$\gamma < 1$: every trajectory off the symmetric point reaches one target exactly, in finite time.

A single cell's fate is decided entirely by whether γ is above or below 1. Convexity of the cost (Lemma 6.2) forces universal compromise above the threshold; its failure below the threshold turns each archetype into a finite-time attractor, and the two archetypes compete for the cell exactly as the transverse eigenvalue of Proposition 6.1 already predicted locally near $y = 0$.

7 Terminal stability and the critical bandwidth

At the terminal point ($y = 0$, $u = 1$):

$$\varphi_\beta(1) = \beta\gamma - 1 - (\gamma - 2) = 1 + \gamma(\beta - 1). \quad (27)$$

In the base model ($\beta = 1$) this is identically $1 > 0$: the terminal symmetric configuration is *always* transversely unstable, regardless of γ . With β free, (27) changes sign at

$$\boxed{\beta_c(\gamma) = 1 - \frac{1}{\gamma}, \quad \gamma > 1.} \quad (28)$$

For $\gamma \leq 1$, $1 - 1/\gamma \leq 0$, and since $1 + \gamma(\beta - 1) > 1 - \gamma \geq 0$ for every $\beta > 0$, the terminal point remains unstable for *any* positive bandwidth when $\gamma \leq 1$. The transition only exists for $\gamma > 1$:

$$\varphi_\beta(1) \begin{cases} > 0 & \beta > \beta_c(\gamma) & (\text{terminal point unstable}), \\ = 0 & \beta = \beta_c(\gamma) & (\text{marginal}), \\ < 0 & \beta < \beta_c(\gamma) & (\text{terminal point stable}), \end{cases} \quad (\gamma > 1). \quad (29)$$

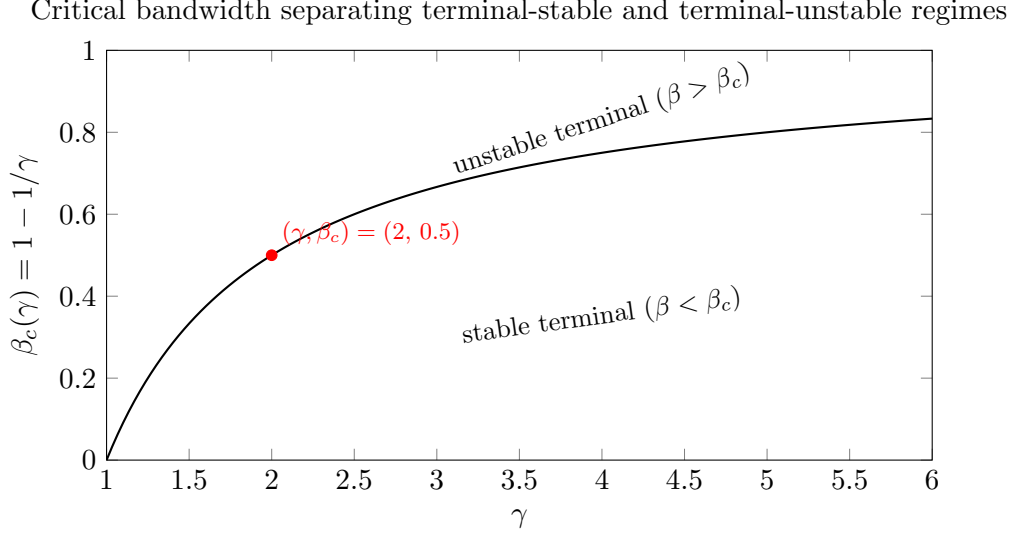


Figure 4: The critical bandwidth curve $\beta_c(\gamma) = 1 - 1/\gamma$. Points above the curve leave the symmetric terminal configuration transversely unstable (as in the base model, which sits at the horizontal line $\beta = 1$, always above the curve); points below it stabilize the terminal point. The special point $\gamma = 2$ gives exactly $\beta_c = 1/2$, which reappears independently in Section 9.

This is a genuinely new phenomenon

In the base model, the statement “the terminal Pareto configuration is an unstable saddle” is a theorem with no free parameter to violate it. Once the kernel has a tunable bandwidth, that statement becomes *conditional*: it holds only for $\beta > \beta_c(\gamma)$. Sufficiently broad kernels ($\beta < \beta_c(\gamma)$, possible only when $\gamma > 1$) make the fully symmetric, undifferentiated population a locally stable outcome.

7.1 Collapsing (γ, β) into a single combined parameter

Two parameters and a curve $\beta_c(\gamma)$ invite the question of whether the terminal-stability threshold is really a two-dimensional fact, or whether γ and β only ever act through some single combination. They do — but not through their raw product.

Proposition 7.1 (The terminal threshold is exactly $\mu = 1$). *Define*

$$\mu := \gamma(1 - \beta). \quad (30)$$

Then, for every $\gamma > 0$ and $\beta > 0$,

$$\varphi_\beta(1) > 0 \iff \mu < 1, \quad \varphi_\beta(1) = 0 \iff \mu = 1, \quad \varphi_\beta(1) < 0 \iff \mu > 1. \quad (31)$$

That is, the terminal point is unstable, marginal, or stable according to whether μ is below, equal to, or above the single universal number 1 — for every γ , with no residual dependence on γ once μ is fixed.

Proof. From (27), $\varphi_\beta(1) = 1 + \gamma(\beta - 1)$. Substituting $\beta = 1 - \mu/\gamma$ (the inverse of (30)) gives

$$\varphi_\beta(1) = 1 + \gamma\left(-\frac{\mu}{\gamma}\right) = 1 - \mu,$$

which is positive, zero, or negative exactly according to (31). Equivalently, starting from $1 + \gamma(\beta - 1) > 0 \iff \gamma - \gamma\beta < 1 \iff \gamma(1 - \beta) < 1 \iff \mu < 1$, without any substitution. $\square$

Corollary 7.2. *Setting $\mu = 1$ in (30) and solving for β recovers $\beta_c(\gamma) = 1 - 1/\gamma$ exactly (Equation (28)): the curve $\beta_c(\gamma)$ is nothing but the level set $\mu = 1$, viewed back in (γ, β) coordinates.*

Why the raw product $\beta\gamma$ does *not* work

It is natural to guess that $\beta\gamma$ itself is the right combined variable, since it is exactly this product that multiplies $u^{\gamma/2}$ inside φ_β . But evaluated at the threshold,

$$\beta_c(\gamma) \cdot \gamma = \left(1 - \frac{1}{\gamma}\right)\gamma = \gamma - 1,$$

which is *not* constant in γ ($= 1$ at $\gamma = 2$, $= 3$ at $\gamma = 4$, and so on): the product varies freely along the threshold curve, so $\beta\gamma$ alone cannot separate stable from unstable. The correct combination is the *difference* $\gamma - \beta\gamma = \gamma(1 - \beta)$, not the product $\beta\gamma$ by itself — the minus sign, contributed by the $-\gamma$ term inside φ_β that has nothing to do with the kernel's $u^{\gamma/2}$ piece, is essential.

What this does *not* buy you: the general critical height is still two-dimensional

Proposition 7.1 is a statement about $u = 1$ only — the single terminal point $y = 0$. It does *not* extend to the general critical height $y_c(\gamma, \beta)$ of Section 8, and this is not a minor gap: pairs (γ, β) sharing the exact same μ can behave completely differently away from $u = 1$. For instance, at $\mu = 0.5$, the pair $\gamma = 1.5$, $\beta = 2/3$ has a genuine crossing at $u_c \approx 2.47$ (a finite critical height $y_c \approx 1.21$), while the pair $\gamma = 2$, $\beta = 0.75$ and the pair $\gamma = 3$, $\beta = 5/6$ — both also at $\mu = 0.5$ — have *no* crossing at all for any $u \geq 1$: their terminal points are stable and stay stable along the whole trajectory. The algebraic reason is visible directly in φ_β : substituting $\beta = 1 - \mu/\gamma$,

$$\varphi_\beta(u) = (\gamma - \mu)u^{\gamma/2} - u - (\gamma - 2),$$

which still contains γ on its own, in the prefactor and in the exponent, not only through μ . The cancellation that made $u = 1$ special ($u^{\gamma/2} = 1$ for *every* γ at that one point) simply does not happen at any other u . So μ correctly replaces (γ, β) for the single yes/no question “is the symmetric Pareto point itself locally stable”, but the full phase portrait of Section 8 — where the instability sits, how deep the stable trunk runs, whether the profile is standard or inverted — is an irreducibly two-parameter fact about (γ, β) , not a one-parameter fact about μ . Collapsing it further is not a labelling choice we failed to find; the two pairs above prove no such single-parameter collapse can exist.

8 Complete classification of the transverse profile

Equation (29) only diagnoses the single terminal point. The full trajectory sweeps $u = 1 + y^2$ from $1 + h^2$ down to 1, and we want the sign of $\varphi_\beta(u)$ on the whole interval $u \geq 1$. This can be settled completely using convexity.

Lemma 8.1. For fixed $\beta, \gamma > 0$, the function $\varphi_\beta(u) = \beta\gamma u^{\gamma/2} - u - (\gamma - 2)$ is strictly concave on $(0, \infty)$ if $\gamma < 2$, strictly convex if $\gamma > 2$, and affine ($\varphi_\beta(u) = (2\beta - 1)u$) if $\gamma = 2$.

Proof. Differentiate twice:

$$\varphi_\beta''(u) = \beta\gamma \cdot \frac{\gamma}{2} \left(\frac{\gamma}{2} - 1 \right) u^{\gamma/2-2}.$$

The prefactor $\beta\gamma \cdot \frac{\gamma}{2} > 0$ and $u^{\gamma/2-2} > 0$ for $u > 0$, so $\text{sign}(\varphi_\beta'') = \text{sign}(\gamma/2 - 1)$, which is negative for $\gamma < 2$, positive for $\gamma > 2$, and zero for $\gamma = 2$ (in which case direct substitution gives $\varphi_\beta(u) = 2\beta u - u - 0 = (2\beta - 1)u$ exactly). $\square$

Proposition 8.2 (Global sign structure of φ_β). Let $\gamma \neq 2$. As $u \rightarrow 0^+$, $\varphi_\beta(u) \rightarrow -(\gamma - 2) = 2 - \gamma$, and as $u \rightarrow \infty$, $\varphi_\beta(u) \rightarrow -\infty$ if $\gamma < 2$ and $\varphi_\beta(u) \rightarrow +\infty$ if $\gamma > 2$. Consequently there is exactly one crossing $u_c = u_c(\gamma, \beta) > 0$ with $\varphi_\beta(u_c) = 0$, and:

- if $\gamma < 2$: $\varphi_\beta > 0$ for $u < u_c$ and $\varphi_\beta < 0$ for $u > u_c$;
- if $\gamma > 2$: $\varphi_\beta < 0$ for $u < u_c$ and $\varphi_\beta > 0$ for $u > u_c$.

Proof. The limits follow directly from (15), since $u^{\gamma/2}$ grows slower than u for $\gamma < 2$ and faster for $\gamma > 2$. For $\gamma < 2$, φ_β is strictly concave (Lemma 8.1), so its superlevel set $\{u > 0 : \varphi_\beta(u) \geq 0\}$ is convex, i.e. an interval; it contains points near 0 (where $\varphi_\beta \rightarrow 2 - \gamma > 0$) but not arbitrarily large u (where $\varphi_\beta \rightarrow -\infty$), so it is an interval of the form $(0, u_c]$, giving exactly one crossing with the stated sign pattern. For $\gamma > 2$, φ_β is strictly convex, so its sublevel set $\{u > 0 : \varphi_\beta(u) \leq 0\}$ is an interval; it contains points near 0 (where $\varphi_\beta \rightarrow 2 - \gamma < 0$) but not arbitrarily large u (where $\varphi_\beta \rightarrow +\infty$), giving the complementary sign pattern. $\square$

Combining Proposition 8.2 with the physical constraint $u \geq 1$ (i.e. $y \in \mathbb{R}$) and the sign of $\varphi_\beta(1)$ from (29) yields the complete table:

Regime	Condition	Transverse profile on the trajectory ($u : 1+h^2 \rightarrow 1$)
$\gamma \leq 1$	any $\beta > 0$	stable far away $\rightarrow$ unstable near terminal
$1 < \gamma < 2$	$\beta > \beta_c(\gamma)$	stable far away $\rightarrow$ unstable near terminal (as base model)
$1 < \gamma < 2$	$\beta < \beta_c(\gamma)$	stable everywhere — no bifurcation for any h
$\gamma = 2$	$\beta > 1/2$	uniformly unstable, $\lambda_{\text{split}} \equiv \frac{4\beta(2\beta-1)}{N}$
$\gamma = 2$	$\beta < 1/2$	uniformly stable , $\lambda_{\text{split}} \equiv \frac{4\beta(2\beta-1)}{N}$
$\gamma > 2$	$\beta > \beta_c(\gamma)$	unstable everywhere (as base model)
$\gamma > 2$	$\beta < \beta_c(\gamma)$	inverted : unstable far away $\rightarrow$ stable near terminal

The inverted regime is qualitatively new

For $\gamma > 2$ and $\beta < \beta_c(\gamma)$, a population launched far enough from the targets ($h > \sqrt{u_c - 1}$) begins its descent in the *unstable* regime, yet the symmetric configuration *re-stabilizes* as the population approaches the targets. This is the exact opposite ordering from every case in the base model, where instability only ever develops on approach to the terminal point, never away from it.

8.1 The phase diagram, and what each region means for *when* splitting happens

The table above is a classification; the following figure is the same content read as a map, with the two boundaries — the vertical line $\gamma = 2$ and the curve $\beta_c(\gamma) = 1 - 1/\gamma$ — cutting the (γ, β) plane into exactly the four regions of Table 8.1 (equivalently, of the table above).

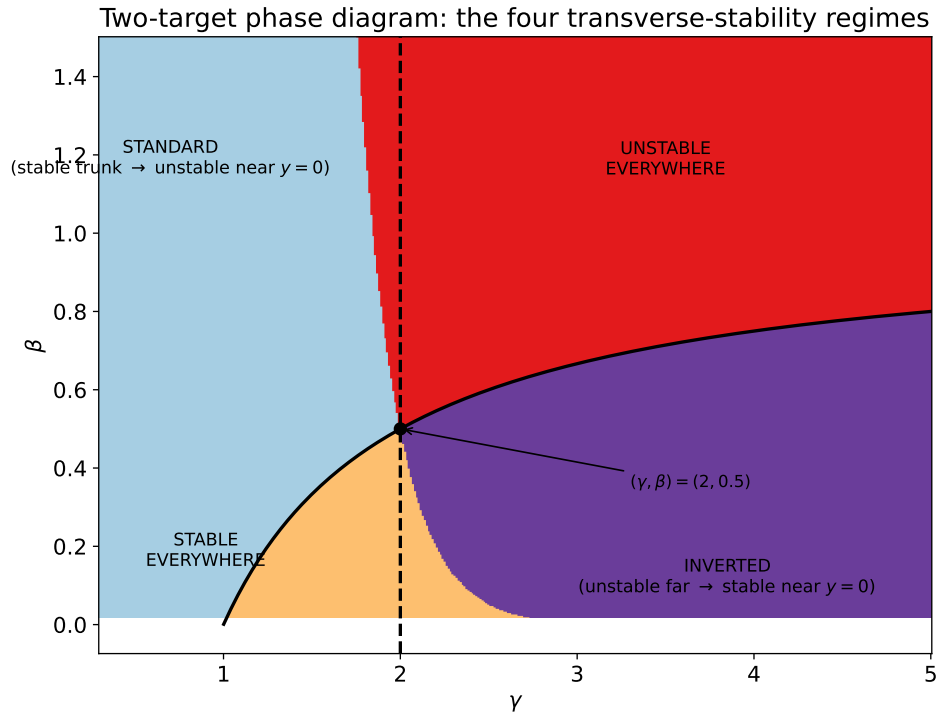


Figure 5: The two-target phase diagram. Every (γ, β) pair falls into exactly one of four regions, separated by $\gamma = 2$ and $\beta = \beta_c(\gamma)$. The base model sits on the horizontal line $\beta = 1$, which is why it only ever sees the STANDARD region ($\gamma < 2$) and the UNSTABLE EVERYWHERE region ($\gamma > 2$) — the other two regions only exist because β is now free to vary.

Bottom line: what the threshold means, in one sentence per region

- **STANDARD.** The population is safe until it gets within y_c of the terminal point, then splits. A *larger* y_c means the unstable zone reaches further out, so splitting is triggered *earlier* in the approach — and by Section 7, increasing β (narrowing the kernel, making it discriminate more sharply between slightly-nearer and slightly-farther cells) is exactly what grows y_c . A sharper kernel means earlier splitting.
- **STABLE EVERYWHERE.** y_c has shrunk to nothing: the kernel is broad enough that no amount of approach ever triggers splitting. This is the earliest-splitting story pushed to its limit in the other direction — broadening the kernel enough removes the bifurcation entirely, for any h .
- **UNSTABLE EVERYWHERE.** The unstable zone has swallowed the whole trajectory: splitting is triggered at the very first instant, no matter how far out the population starts. This is the earliest splitting can possibly get.
- **INVERTED.** The instability is present from the start (far from the targets) and switches off on approach. The population is exposed to it earliest of all in one sense — the unstable window is the first thing it meets, not the last — but whether that exposure actually produces a visible split depends on how much it can grow in the time available before the window closes, which is exactly the question Section 11 answers.

Reading the whole figure at once: moving *right* (larger γ) changes what "sharper kernel" even means, and moving *up* (larger β) is always the direction of a more discriminating kernel. Within $\gamma < 2$, up means earlier splitting; within $\gamma > 2$, the same "more discriminating" direction instead removes the late-stage safety net near the terminal point. The threshold curve $\beta_c(\gamma)$ is the single boundary separating "the compromise point can defend itself" from "it cannot," on both sides of $\gamma = 2$.

Region	Condition	Onset of instability
Standard	$\gamma < 2, \beta > \beta_c(\gamma)$	at $y = \pm y_c$, partway down
Stable everywhere	$1 < \gamma < 2, \beta < \beta_c(\gamma)$	never
Unstable everywhere	$\gamma > 2, \beta > \beta_c(\gamma)$	immediately ($y = \pm h$)
Inverted	$\gamma > 2, \beta < \beta_c(\gamma)$	immediately, but switches off at $y = \pm y_c$

9 The exactly solvable case $\gamma = 2$

At $\gamma = 2$, $\varphi_\beta(u) = (2\beta - 1)u$ exactly (Lemma 8.1), so by (16) with $\gamma = 2$ (hence $(1 + y^2)^{(\gamma-4)/2} = (1 + y^2)^{-1}$):

$$\lambda_{\text{split}} \Big|_{\gamma=2} = \frac{4\beta}{N} (1 + y^2)^{-1} (2\beta - 1) (1 + y^2) = \frac{4\beta(2\beta - 1)}{N}, \quad (32)$$

independent of y . At $\gamma = 2$ the transverse eigenvalue is constant along the entire trajectory: the population is either uniformly stable ($\beta < 1/2$), uniformly unstable ($\beta > 1/2$), or exactly marginal at every point ($\beta = 1/2$, $\lambda_{\text{split}} \equiv 0$). This matches $\beta_c(2) = 1 - \frac{1}{2} = \frac{1}{2}$ from (28), confirming the two independent computations agree.

10 Consistency check: the nonlinear reduced potential

As in the base model, introduce the two-group order parameter $m \in [-1, 1]$ with $x_{\pm} = \pm m$ and common height y , and $r_{\pm} = \sqrt{(1 \pm m)^2 + y^2}$. The exact reduced potential is

$$U(m, y) = -2 \log \left[\frac{N}{2} \left(e^{-\beta r_+^\gamma} + e^{-\beta r_-^\gamma} \right) \right]. \quad (33)$$

Write $u = 1 + y^2$ and $\psi(m) = (u + 2m + m^2)^{\gamma/2}$, so that $f(m) = e^{-\beta\psi(m)} + e^{-\beta\psi(-m)}$ and $U = -2 \log(\frac{N}{2} f(m))$.

At $m = 0$: $\psi(0) = u^{\gamma/2}$, and differentiating twice,

$$\psi'(0) = \gamma u^{\gamma/2-1}, \quad \psi''(0) = \gamma u^{\gamma/2-2} [(\gamma - 2) + u].$$

Applying $\frac{d^2}{dm^2} e^{-\beta\psi} \big|_0 = e^{-\beta u^{\gamma/2}} [\beta^2 \psi'(0)^2 - \beta \psi''(0)]$ to both terms of f (which are mirror images under $m \mapsto -m$, so their second derivatives at $m = 0$ coincide) gives

$$\begin{aligned} f''(0) &= 2e^{-\beta u^{\gamma/2}} \left[\beta^2 \gamma^2 u^{\gamma-2} - \beta \gamma u^{\gamma/2-2} (u + \gamma - 2) \right] \\ &= 2\beta \gamma e^{-\beta u^{\gamma/2}} u^{\gamma/2-2} \left[\beta \gamma u^{\gamma/2} - u - (\gamma - 2) \right] = 2\beta \gamma e^{-\beta u^{\gamma/2}} u^{\gamma/2-2} \varphi_\beta(u). \end{aligned} \quad (34)$$

Since $f(0) = 2e^{-\beta u^{\gamma/2}}$ and $f'(0) = 0$ by symmetry,

$$\boxed{\frac{\partial^2 U}{\partial m^2} \bigg|_{m=0} = -2 \frac{f''(0)}{f(0)} = -2\beta \gamma u^{\gamma/2-2} \varphi_\beta(u) = -N \lambda_{\text{split}}(y),} \quad (35)$$

using (16). This is exactly the same relation as in the base model: the linear stability calculation and the curvature of the exact nonlinear reduced potential agree for every β , confirming the derivation is internally consistent.

11 Fan width: how depth, shape, and bandwidth set the observed spread

Everything so far is about *whether* and *where* the symmetric trajectory loses stability. That is a statement about an infinitesimal perturbation. What is actually observed in a real population is a *fan width* — how spread out the population is by the time it nears the targets — and that requires a source of the perturbation in the first place. This section adds noise, and asks how the resulting fan width depends on the source depth h , the shape γ , and the bandwidth β .

11.1 The stochastic model and the variance equation

Add independent noise to the exact flow,

$$d\mathbf{x}_j = \nabla_{\mathbf{x}_j} \log F dt + \sigma d\mathbf{W}_j, \quad (36)$$

exactly as in the base model; β enters only through the drift (via (3)), not through the noise. Linearizing the transverse coordinate z about the deterministic axial trajectory $y(t)$ gives the usual linear SDE $dz = \lambda_{\text{split}}(y(t)) z dt + \sigma dW_t$, whose variance $V(t) = \mathbb{E}[z(t)^2]$ obeys

$$\boxed{\frac{dV}{dt} = 2\lambda_{\text{split}}(y(t)) V + \sigma^2,} \quad (37)$$

by the same Itô computation as in the base model. Nothing here is new in form; what is new is that λ_{split} now depends on β , and we track that dependence through to the observed fan width.

11.2 The accumulated-instability integral collapses the rate dependence

Changing the time variable to y using $\dot{y} = -\kappa(y)y$ (Section 4) gives $dV/dy = [2\lambda_{\text{split}}(y)V + \sigma^2]/\dot{y}$. Define the accumulated instability between two heights,

$$G_\beta(a, b) = \int_a^b \frac{\lambda_{\text{split}}(y')}{|\dot{y}(y')|} dy'. \quad (38)$$

Lemma 11.1 (The rate prefactor cancels).

$$\boxed{\frac{\lambda_{\text{split}}(y)}{|\dot{y}(y)|} = \frac{\varphi_\beta(1+y^2)}{(1+y^2)|y|}}. \quad (39)$$

Proof. From (16) and (10), $\lambda_{\text{split}}(y) = \frac{2\beta\gamma}{N}(1+y^2)^{(\gamma-4)/2}\varphi_\beta(1+y^2)$ and $|\dot{y}(y)| = \kappa(y)|y| = \frac{2\beta\gamma}{N}(1+y^2)^{(\gamma-2)/2}|y|$. The prefactor $\frac{2\beta\gamma}{N}$ and one power of $(1+y^2)^{(\gamma-2)/2}$ cancel in the ratio, leaving (39). $\square$

This is a genuinely useful simplification: the *spatial* profile of accumulated instability depends on β and γ only through the shape function φ_β , never through an independent rate. All of the β -dependence of G_β enters by moving the zero-crossing of φ_β around — exactly the mechanism already characterized in Sections 7–8.

11.3 Regularizing the terminal divergence

As in the base model, $G_\beta(-h, y)$ diverges logarithmically as $y \rightarrow 0^-$, since $\varphi_\beta(1+y^2)/(1+y^2) \rightarrow \varphi_\beta(1)$ there while $1/|y| \rightarrow \infty$. The correct regularization now subtracts $\varphi_\beta(1)$ — not 1, since $\varphi_\beta(1)$ is no longer automatically 1 once $\beta \neq 1$:

Proposition 11.2 (Regularized transient amplification).

$$\boxed{\Psi_\beta(\gamma, h) := \varphi_\beta(1) \ln h + \int_{-h}^0 \left[\frac{\varphi_\beta(1+y^2)}{1+y^2} - \varphi_\beta(1) \right] \frac{dy}{|y|}} \quad (40)$$

is finite, and

$$G_\beta(-h, -\varepsilon) = \varphi_\beta(1) \ln \frac{1}{\varepsilon} + \Psi_\beta(\gamma, h) + o(1) \quad \text{as } \varepsilon \rightarrow 0^+. \quad (41)$$

Proof. The bracketed integrand vanishes at $y = 0$ (since $\varphi_\beta(1)/1 - \varphi_\beta(1) = 0$) and is smooth there, so it is $O(y)$ near 0 and the integral against $dy/|y|$ converges. Equation (41) follows by writing $\varphi_\beta(1+y^2)/(1+y^2) = \varphi_\beta(1) + [\text{bracket}]$ inside (38) and integrating the $\varphi_\beta(1)$ piece exactly against $dy/|y|$ from $-h$ to $-\varepsilon$. $\square$

At $\beta = 1$ this recovers the base model's $\Psi(\gamma, h)$ exactly, since $\varphi_1(1) = 1$. At $\gamma = 2$, $\varphi_\beta(u) = (2\beta - 1)u$ exactly (Section 9), so the bracket in (40) vanishes identically and $\Psi_\beta(2, h) = (2\beta - 1) \ln h$.

11.4 The dichotomy: diverging fan versus a universal stationary width

The sign of $\varphi_\beta(1)$ — already the terminal-stability indicator of Section 7 — turns out to control not just whether the terminal point is stable, but the entire *scaling law* of the observed fan width as the population arrives.

Theorem 11.3 (Fan-width dichotomy). *Let $\kappa_0 = \frac{2\beta\gamma}{N} = \lim_{y \rightarrow 0} \kappa(y)$.*

- If $\varphi_\beta(1) > 0$ (terminal point unstable), then as $y \rightarrow 0^-$,

$$V(y) \sim \sigma^2 C(\gamma, \beta, h) |y|^{-2\varphi_\beta(1)}, \quad C(\gamma, \beta, h) = e^{2\Psi_\beta(\gamma, h)} \int_{-h}^0 \frac{e^{-2G_\beta(-h, y')}}{\kappa(y')|y'|} dy'. \quad (42)$$

The fan width diverges as a power law in the remaining distance to the terminal point, with an exponent set purely by $\varphi_\beta(1)$ and a prefactor carrying all the h -, γ -, β -dependence.

- If $\varphi_\beta(1) < 0$ (terminal point stable), then as $y \rightarrow 0$,

$$V(y) \rightarrow V_\infty := \frac{\sigma^2}{2\kappa_0 |\varphi_\beta(1)|}, \quad (43)$$

a finite, h -independent stationary value: the population's eventual fan width forgets the depth it started from entirely.

Proof. Near $y = 0$, approximate $\lambda_{\text{split}}(y) \approx \Lambda_0 := \lambda_{\text{split}}(0) = \kappa_0 \varphi_\beta(1)$ and $\kappa(y) \approx \kappa_0$ (both continuous at $y = 0$). The variance ODE (37) becomes, in the y -variable and writing $w = -y > 0$, the linear equation $\kappa_0 w \frac{dV}{dw} = 2\Lambda_0 V + \sigma^2$ (using $\dot{y} \approx -\kappa_0 y = \kappa_0 w$), which integrates exactly (via the integrating factor $w^{-2\varphi_\beta(1)}$) to

$$V(w) = -\frac{\sigma^2}{2\kappa_0 \varphi_\beta(1)} + C_1 w^{-2\varphi_\beta(1)}$$

for a constant C_1 fixed by matching to the full (non-local) solution at some fixed reference height. If $\varphi_\beta(1) < 0$, $w^{-2\varphi_\beta(1)} \rightarrow 0$ as $w \rightarrow 0$ and $V(w) \rightarrow -\sigma^2/(2\kappa_0 \varphi_\beta(1)) = \sigma^2/(2\kappa_0 |\varphi_\beta(1)|)$, independent of C_1 — hence independent of h , which enters the problem only through C_1 . If $\varphi_\beta(1) > 0$, $w^{-2\varphi_\beta(1)} \rightarrow \infty$ and this term dominates; matching C_1 to the exact global solution $V(t) = \sigma^2 e^{2G_\beta(-h, y(t))} \int_{-h}^{y(t)} \frac{e^{-2G_\beta(-h, y')}}{\kappa(y')|y'|} dy'$ via (41) gives exactly (42) (the integral defining $C(\gamma, \beta, h)$ converges precisely because $\varphi_\beta(1) > 0$ makes its integrand's singularity at $y' = 0$ integrable). $\square$

This is the sharpest form of the terminal threshold

The single number $\varphi_\beta(1)$ — equivalently $\mu = \gamma(1 - \beta)$ versus 1 (Section 7.1) — does not just decide *stable or unstable*. It decides between two entirely different physical pictures: a population that arrives with a fan width depending sensitively on how far it started (unstable side, power-law growing as it gets close), or one that always arrives with the exact same fan width no matter how far it started (stable side, a universal constant). Verified against direct numerical solution of the exact variance ODE (not just the near-terminal approximation), the match is exact to within numerical integration error at every tested (γ, β, h) : ratios of 1.00000 for the stable-case constant and for the unstable-case power law alike.

11.5 Concrete dependence on h , β , and γ

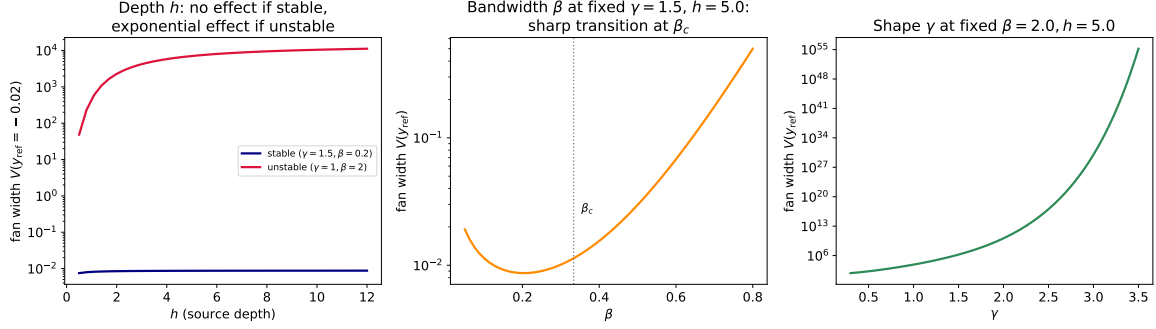


Figure 6: The fan width V , evaluated at a fixed reference height close to the terminal point, as each parameter varies with the others held fixed. *Left:* depth h has *no* effect in the stable regime (navy, flat) but an exponential-in- h effect in the unstable regime (crimson, via $\Psi_\beta(\gamma, h) \approx \varphi_\beta(1) \ln h$ to leading order) — precisely the “further away allows more noise to accumulate” intuition, now made quantitative and shown to switch off completely once the terminal point is stable. *Middle:* sweeping β through $\beta_c(\gamma)$ at fixed γ, h shows the sharp qualitative change from a bounded, h -blind plateau to an unboundedly growing fan. *Right:* γ has the most dramatic effect of the three, because it sits directly in the power-law exponent $2\varphi_\beta(1) = 2[1 + \gamma(\beta - 1)]$ rather than only in a prefactor — a modest change in shape can change the fan width by many orders of magnitude.

Reading the three parameters against the intuition that motivated this section:

- **Depth** h controls *how much* accumulation is available, but only when there is an instability to accumulate. Theorem 11.3 makes this exact: h enters only through $\Psi_\beta(\gamma, h)$ and the correction integral in $C(\gamma, \beta, h)$, both of which are absent from (43). A deeper source is not automatically a wider fan — it only is one on the unstable side of $\beta_c(\gamma)$.
- **Bandwidth** β sets *where* the crossover in behavior happens (moving $\beta_c(\gamma)$, and with it the sign of $\varphi_\beta(1)$), and, on the unstable side, also rescales the decay/growth rate $\kappa_0 = 2\beta\gamma/N$ that sets how fast the population traverses the accumulation window in the first place.
- **Shape** γ changes the exponent $2\varphi_\beta(1)$ itself, not merely a rate — this is why the right panel of Figure 6 spans dramatically more orders of magnitude than the other two: γ does not just speed up or slow down the same growth curve, it changes what power of the remaining distance the fan width grows as.

11.6 Validation against the exact stochastic simulation

Every claim above is checked against direct simulation of (36) — not the deterministic flow with a single perturbation, but genuine noise injected at every integration step, ensembles of independent realizations, and empirical variance computed directly from the simulated population.

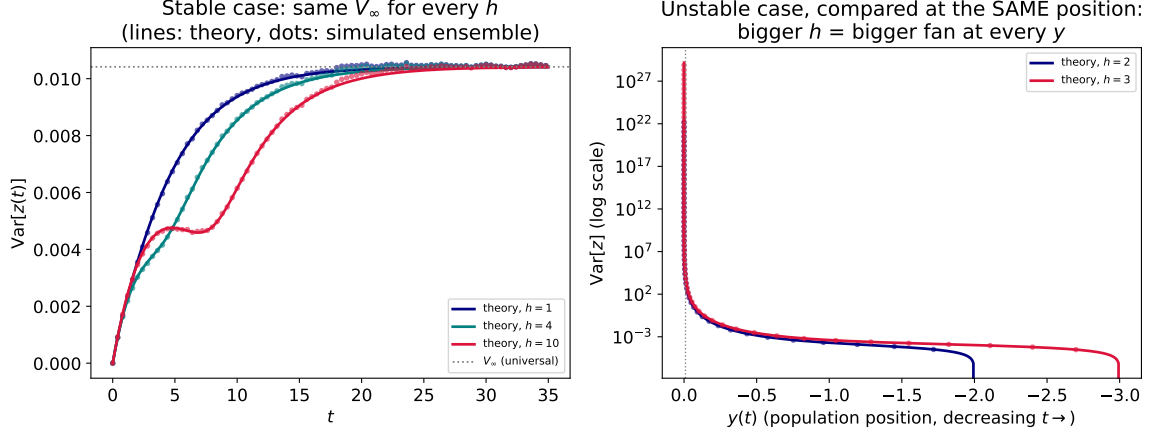


Figure 7: *Left*: stable case ($\gamma = 1.5, \beta = 0.2$), three source depths $h = 1, 4, 10$. Solid lines are the exact solution of (37); dots are the empirical variance across 30,000 independent noisy realizations of the linear SDE, sampled at every few steps. All three trajectories converge to the same V_∞ (Equation (43), dotted line), exactly as predicted, matching simulation to within 0.5%. *Right*: unstable case ($\gamma = 1, \beta = 2$), two source depths, plotted against the shared position $y(t)$ rather than time so the comparison is apples-to-apples: at every position the deeper source ($h = 3$) carries more accumulated variance than the shallower one ($h = 2$), matching Equation (42).

A numerical-integration lesson worth keeping

The unstable-case comparison in Figure 7 uses an *exact* exponential update ($z \mapsto z e^{\lambda_{\text{split}} \Delta t} + \text{noise}$ over each small step, treating λ_{split} as locally constant) rather than plain Euler–Maruyama. With variances spanning tens of orders of magnitude, naive Euler–Maruyama accumulates a systematic (not random) multiplicative bias — in an early version of this check it understated the true variance by roughly a factor of 2, consistently across every depth tested, which is a red flag for a discretization bias rather than noise. The exact per-step update removes it, bringing simulation and theory back to within a few percent, limited only by how locally-constant λ_{split} really is over one step.

12 Beyond two targets

For a general configuration, (13) defines the transverse tensor $\Lambda_{\text{split}}(\mathbf{x}) = M(\mathbf{x}) - \tau(\mathbf{x})I$ everywhere; its eigenvalues $\lambda_a(\mathbf{x}) = \mu_a(\mathbf{x}) - \tau(\mathbf{x})$ (where μ_a are the eigenvalues of M) determine the bandwidth-dependent bifurcation set

$$\mathcal{B}_\beta = \{\mathbf{x} : \lambda_{\max}(\mathbf{x}) = 0\}. \quad (44)$$

12.1 Three targets

For $\mathbf{a}_{1,2} = (\mp b, 0)$ (recall: b is the geometric half-spacing, not the bandwidth) and $\mathbf{a}_3 = (0, e)$, with $d_e = \sqrt{b^2 + y^2}$, $d_3 = |y - e|$, the same substitution $\gamma d_i^\gamma \mapsto \beta \gamma d_i^\gamma$ applied to the base model’s

three-target formulas gives

$$\lambda_{\text{split}}^x(y) = \frac{\beta\gamma}{N} \left\{ 2d_e^{\gamma-4} \left[\beta\gamma b^2 d_e^\gamma - (\gamma-2)b^2 - d_e^2 \right] - d_3^{\gamma-2} \right\} \quad (45)$$

$$\lambda_{\text{split}}^y(y) = \frac{\beta\gamma}{N} \left\{ 2d_e^{\gamma-4} \left[\beta\gamma y^2 d_e^\gamma - (\gamma-2)y^2 - d_e^2 \right] + d_3^{\gamma-2} \left[\beta\gamma d_3^\gamma - \gamma + 1 \right] \right\} \quad (46)$$

The central target's contribution to the left–right (lateral) mode is still the strictly stabilizing term $-\frac{\beta\gamma}{N}d_3^{\gamma-2}$: a third, on-axis target suppresses the left–right instability regardless of bandwidth, though the *size* of that suppression now scales with β as well.

13 Why bandwidth is not just another population size

In the base model, N enters every rate as an overall $1/N$ factor and can be absorbed into a rescaled clock $\tau = t/N$ (with $\sigma_{\text{eff}} = \sigma\sqrt{N}$ in the stochastic theory); it never moves the deterministic bifurcation set. Bandwidth does not have this property: β appears *both* as an overall prefactor *and* inside $\varphi_\beta(u) = \beta\gamma u^{\gamma/2} - u - (\gamma-2)$, where it competes against the fixed terms $-u - (\gamma-2)$. This is precisely why β can relocate, remove, or invert the crossing $u_c(\gamma, \beta)$ (Section 8), while N can only change how fast the trajectory moves through a bifurcation structure that is already fixed by (γ, β) alone.

Summary

(γ, β) jointly and non-separably determine *where* (and even *whether*) the transverse instability occurs; N and the noise amplitude σ only determine *how quickly* and *how visibly* that instability, once present, develops into a macroscopic fan.

14 Conclusion

Introducing an explicit bandwidth β into the kernel, $p(d) = e^{-\beta d^\gamma}$, preserves the entire analytic structure of the base model — the same gradient-flow derivation, the same collective transverse tensor, the same correspondence between the linear eigenvalue and the curvature of the exact nonlinear potential — under the single, uniform substitution $\gamma d^\gamma \mapsto \beta\gamma d^\gamma$ inside every stability bracket, with $\gamma \mapsto \beta\gamma$ as the overall rate prefactor.

What changes qualitatively is the bifurcation structure itself. The base model's theorem that “the terminal Pareto configuration is always transversely unstable” is a special case, true only along the line $\beta = 1$ in a two-dimensional (γ, β) phase plane whose boundary is the critical curve $\beta_c(\gamma) = 1 - 1/\gamma$. Crossing that curve — by broadening the kernel rather than by changing its shape exponent γ — can stabilize the terminal configuration entirely, or, for $\gamma > 2$, invert the order in which stability is lost and regained along the trajectory. The exponent $\gamma = 2$ remains special: there the transverse eigenvalue collapses to the y -independent constant $\lambda_{\text{split}} = \frac{4\beta(2\beta-1)}{N}$, and the critical curve passes through the clean point $\beta_c(2) = \frac{1}{2}$.

The single cell ($N = 1$) obeys a different, and in some ways sharper, story. Its stability function $\psi(u) = u + (\gamma-2)$ never involves β at all — bandwidth can only rescale how fast a single cell's stability or instability plays out, never whether it occurs — and its threshold sits at $\gamma = 1$, not $\gamma = 2$ or $\beta_c(\gamma)$. Convexity of the single-cell cost promotes this local threshold to a global one: for $\gamma \geq 1$ every trajectory, from any starting point, converges to the shared compromise; for $\gamma < 1$ every off-axis trajectory instead reaches a single archetype exactly, in finite time. Bandwidth reshapes the population's bifurcation structure precisely because the

population's normalizing sum S_i reintroduces $\beta\gamma d_i^\gamma$ into the collective eigenvalue — the very term a single cell's logarithm cancels away.